NewStart AI — Government Document Routing (Demo)

Research Results   Historical Random Form Demo   Family-Aware Robustness Demo

Model comparison (frozen test set)

Method	Accuracy	Macro F1	Weighted F1	Mean latency (ms)	Cost / document
bert	0.9934	0.9741	0.9936	16.4	free (local, after training)
llm	0.9868	0.9694	0.9870	1507.9	$0.000728
llm_rag	0.9868	0.9694	0.9870	1555.8	$0.000815

Per-class F1

Label	Method	Precision	Recall	F1	Support
USCIS	bert	1.000	1.000	1.000	51
DMV	bert	1.000	1.000	1.000	55
SSA	bert	1.000	0.975	0.987	40
IRS	bert	0.833	1.000	0.909	5
USCIS	llm	0.981	1.000	0.990	51
DMV	llm	1.000	0.982	0.991	55
SSA	llm	1.000	0.975	0.987	40

Label	Method	Precision	Recall	F1	Support
IRS	llm	0.833	1.000	0.909	5
USCIS	llm_rag	0.981	1.000	0.990	51
DMV	llm_rag	1.000	0.982	0.991	55
SSA	llm_rag	1.000	0.975	0.987	40
IRS	llm_rag	0.833	1.000	0.909	5

IRS support is only ~5 test documents — its precision/recall/F1 should be read as directional, not precise.

Confusion matrices

BERT

	USCIS	DMV	SSA	IRS
USCIS	51	0	0	0
DMV	0	55	0	0
SSA	0	0	39	1
IRS	0	0	0	5

LLM

	USCIS	DMV	SSA	IRS
USCIS	51	0	0	0
DMV	1	54	0	0

	USCIS	DMV	SSA	IRS
SSA	0	0	39	1
IRS	0	0	0	5

LLM_RAG

	USCIS	DMV	SSA	IRS
USCIS	51	0	0	0
DMV	1	54	0	0
SSA	0	0	39	1
IRS	0	0	0	5

Error analysis

Document	True label	BERT	LLM	LLM+RAG	Note
533	DMV	DMV	USCIS	USCIS	Labeled DMV but extracted text is an empty PDF-portfolio placeholder -- an upstream extraction failure. LLM and LLM+RAG guessed USCIS with no real signal; BERT happened to predict DMV correctly.
541	SSA	IRS	IRS	IRS	Labeled SSA but text is IRS Form SS-4 (EIN application) -- likely a dataset labeling error, not a model failure. All three methods predicted IRS.

Limitations

- IRS test-set size (~5 documents): all IRS per-class metrics are statistically noisy and should be read as directional, not precise.
- Two dataset-quality issues surfaced by error analysis (one likely mislabel, one empty PDF extraction) -- not model weaknesses. See the error analysis below.
- All three methods scored macro F1 >= 0.97 and tied at validation macro F1 = 1.000; the dataset's four agencies are lexically very distinct, so this result may not generalize to messier, more ambiguous real-world documents.
- LLM+RAG made identical predictions to the plain LLM on this test set -- retrieval context showed no measurable benefit here.

Reproducibility manifest


```
{
  "dataset_fingerprint": "dcbdbbcb287b203c1faae8a4b3583980d92149e116d3ad87f86cb5bab4ced3f3",
  "split": {
    "random_seed": 42,
    "train_rows": 482,
    "validation_rows": 121,
    "test_rows": 151
  },
  "bert": {
    "artifact_id": "3628681550d7433b94407f684946bb2f",
    "base_model": "bert-base-uncased",
    "long_document_strategy": "first_512",
    "best_validation_macro_f1": 1
  },
  "llm": {
    "provider": "gemini",
    "model": "gemini-3.6-flash",
    "classification_prompt_version": "v1"
  },
  "rag": {
    "embedding_model": "gemini-embedding-001",
```

```
    "vector_store": "chromadb",
    "top_k": 5
  },
  "demo_default_routing_method": "bert",
  "environment": {
    "python_version": "3.11.15",
    "platform": "Windows-10-10.0.26200-SP0"
  }
}
```


Family-Aware Robustness Study (Version 6)

This section is a separate, stricter, leakage-controlled evaluation on a different 99-document test set (drawn only from effective families entirely unseen during training) than the historical results above. ***Do not describe the family-aware model as having 'improved' or 'become worse' than the historical model based on comparing these two raw scores -- the test sets differ in size, composition, and leakage guarantees. The historical score is reported here as context only. The valid head-to-head comparison is the new BERT vs. LLM vs. LLM+RAG evaluation on these same 99 test documents and identical condition inputs, once that comparison is run.***

Primary result — `complete_unmasked` (n=99)

Method	Macro F1	Accuracy	Macro precision	Macro recall
bert	1.0000	1.0000	1.0000	1.0000
llm	1.0000	1.0000	1.0000	1.0000
llm_rag	1.0000	1.0000	1.0000	1.0000

Robustness across all ten conditions

- Family-aware BERT: perfect on **5/10** conditions

- Plain Gemini LLM: perfect on **10/10** conditions
- Gemini LLM+RAG: perfect on **10/10** conditions
- RAG changed **0/990** predictions relative to the plain LLM
- IRS test support: **4 documents only** — treat any IRS-specific number as high-variance, not precise

Condition	Masked?	BERT Macro F1	Plain LLM Macro F1	LLM+RAG Macro F1
complete_unmasked	no	1.0000	1.0000	1.0000
beginning_only_unmasked	no	0.9587	1.0000	1.0000
middle_only_unmasked	no	1.0000	1.0000	1.0000
end_only_unmasked	no	0.9910	1.0000	1.0000
beginning_middle_end_unmasked	no	1.0000	1.0000	1.0000
complete_masked	yes	0.9664	1.0000	1.0000
beginning_only_masked	yes	0.9664	1.0000	1.0000
middle_only_masked	yes	1.0000	1.0000	1.0000
end_only_masked	yes	0.9842	1.0000	1.0000
beginning_middle_end_masked	yes	1.0000	1.0000	1.0000

BERT’s robustness errors (4 unique documents)

Document	Effective family	True label	Failing conditions → predicted
542	IRS:W4V	IRS	beginning_only_unmasked → SSA
634	SSA:SSA3373BK	SSA	end_only_unmasked → DMV, end_only_masked → DMV
601	SSA:SSA1694	SSA	complete_masked → IRS, beginning_only_masked → IRS

Document	Effective family	True label	Failing conditions → predicted
68	USCIS:N565	USCIS	end_only_masked → DMV

Cost and latency

Method	Cases	Prompt tokens	Completion tokens	Cost (USD)	Basis	Mean latency (ms)
Plain Gemini LLM	990	871634	8577	$0.145656	API-confirmed	1647.9
Gemini LLM+RAG	990	3620336	9791	$0.431007	API-confirmed	1783.1
Query embeddings (LLM+RAG)	823	–	–	$0.112684	estimated (token count not API-confirmed)	–

Total: $0.6893 (a calculated estimate). This total is a calculated estimate: API-confirmed generation token usage (plain LLM + LLM+RAG classification calls, via Gemini's response.usage_metadata) plus an estimated query-embedding token count at a documented placeholder rate ($0.15 per million tokens) -- the Gemini Developer API does not return confirmed token counts for embedding calls. Not a fully billed or entirely API-confirmed figure.

> *Plain Gemini LLM and Gemini LLM+RAG both scored a perfect Macro F1/accuracy of 1.000 on every one of the ten frozen conditions in this evaluation -- a ceiling result for this specific 99-document, four-agency test set, not a general claim that LLMs are more robust than BERT. RAG changed zero predictions relative to the plain LLM (in either direction) across all 990 evaluated cases, consistent with a ceiling effect (nothing left to correct) rather than evidence that retrieval context has no value in general.*